



Modelling Financial Markets

Seminar 19: Review and Project Workshop — Hands-on



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Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- AR: AutoRegressive (model)
- BET: Bucharest Exchange Trading (index)
- BH: Benjamini–Hochberg (procedure)
- BIC: Bayesian Information Criterion
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CI: Confidence Interval
- COVID-19: COronaVirus Disease 2019
- DM: Diebold–Mariano (test)
- DQ: Dynamic Quantile (test)
- ES: Expected Shortfall
- ETF: Exchange-Traded Fund
- FDR: False Discovery Rate
- FHS: Filtered Historical Simulation
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- FWER: Family-Wise Error Rate
- FZ0: Fissler–Ziegel loss, zero-homogeneous version
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HS: Historical Simulation
- IGARCH: Integrated GARCH
- IQR: InterQuartile Range
- MCS: Model Confidence Set

Acronyms used in this material (2/2)

- QMLE: Quasi-Maximum Likelihood Estimator
- SE: Standard Error
- WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Introduction

Part A: Derivations Across the Course

Part B: Integrated Case Study

Part C: The Project Proposal

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the key inference results of the course: QMLE sandwich, delta method, power of backtests, multiple testing, tail indices
- ▣ Run the full chain on one market with valid inference: tail index, boundary-aware GARCH interval, DQ, FZ0 and the model confidence set
- ▣ Spot the look-ahead and calendar pitfalls in your own code
- ▣ Write a one-page project proposal with identification, power and a specification grid

- ▣ Background: Lecture 19 and the lectures it reviews; textbooks [Franke, Härdle & Hafner \(2019\)](#) and [McNeil, Frey & Embrechts \(2015\)](#)

Structure

- ▣ **Part A:** twelve derivations across the course (A1–A12)
- ▣ **Part B:** an integrated case study on BET and Bitcoin, with inference (B1–B8)
- ▣ **Part C:** the project proposal: template, worked example, your proposal (C1); critique of an AI-written backtest (C2); a project seed on nonstandard errors (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every exercise gives the data and assumptions, numbered tasks (a), (b), ..., what to report and interpretation questions; solutions go step by step
- Sign convention: loss $L = -r$ (the negative daily log return, in %); VaR and ES are positive; the level is the tail probability (VaR 1%, ES 2.5%)
- Code: [seminar notebook \(Colab\)](#), built on the Quantlet MFM_ch19_seminar

Solved and Proposed Problems

Solved: worked examples in this deck

- ▣ Part A: A1 (QMLE sandwich), A3 (GARCH forecast), A5 (Normal VaR and ES), A7 (power of Kupiec), A9 (Holm and BH), A11 (kurtosis with $\alpha < 4$)
- ▣ Part B: B1 (tail index and robust portmanteau, BET), B3 (profile interval for GARCH persistence), B5 (VaR and ES backtests, DQ, FZ0, MCS), B7 (the look-ahead trap)
- ▣ Part C: a worked project proposal

Proposed: you solve them

- ▣ Part A: A2 (model A1), A4 (model A3), A6 (model A5), A8 (model A7), A10 (model A9), A12 (model A11)
- ▣ Part B: B2 (model B1), B4 (model B3), B6 (model B5), B8 (model B7)
- ▣ Part C: C1, your team's proposal; C2, critique an AI-written backtest (model: B5, B7); C3, nonstandard errors in a mini-multiverse (model: A9)

A1. The QMLE Sandwich for GARCH(1,1) [Solved]

Task

□ Data and assumptions

- ▶ Model: $r_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$, $\theta = (\omega, \alpha, \beta)^\top$; z_t i.i.d. with $\mathbb{E}z_t = 0$, $\mathbb{E}z_t^2 = 1$, $\kappa = \mathbb{E}z_t^4 < \infty$.
- ▶ Estimation by the Gaussian quasi-likelihood $\ell_t(\theta) = -\frac{1}{2}(\ln \sigma_t^2 + r_t^2/\sigma_t^2)$; notation $g_t = \sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$, known at $t-1$ [Bollerslev & Wooldridge \(1992\)](#).
- ▶ For (c): the GARCH(1,1)- t fit to BET in B3 gives $\hat{\nu} = 5.011$; for Student- t innovations $\kappa = 3(\nu - 2)/(\nu - 4)$ when $\nu > 4$.

□ Tasks

- ▶ (a) Differentiate ℓ_t with respect to θ , write the score $s_t = \partial \ell_t / \partial \theta$ in terms of z_t and g_t , and show that $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] = 0$.
- ▶ (b) Compute $A = -\mathbb{E}[\partial s_t / \partial \theta^\top]$ and $B = \mathbb{E}[s_t s_t^\top]$ using $\mathbb{E}z_t^2 = 1$ and $\mathbb{E}z_t^4 = \kappa$, and show that the sandwich equals $A^{-1} B A^{-1} = \frac{\kappa-1}{2} A^{-1}$.
- ▶ (c) Substitute $\hat{\nu} = 5.011$: compute κ , the correction factor $(\kappa - 1)/2$ for variances and $\sqrt{(\kappa - 1)/2}$ for standard errors.

□ Report

- ▶ The expressions of s_t , A and B ; the values of κ and of both correction factors.

□ Interpretation question

- ▶ When are the Hessian-only standard errors A^{-1} correct, by how much do they inflate the t -statistics for BET, and what happens when $\nu \leq 4$?

A1. The QMLE Sandwich for GARCH(1,1) (Solution 1/2) [Solved]

- (a) Step 1: $\partial \ell_t / \partial \sigma_t^2 = -\frac{1}{2} \left(\frac{1}{\sigma_t^2} - \frac{r_t^2}{\sigma_t^4} \right) = \frac{1}{2\sigma_t^2} (z_t^2 - 1)$.
- Step 2 (chain rule): $s_t = \frac{1}{2}(z_t^2 - 1) g_t$; g_t is known at $t - 1$ and $\mathbb{E}[z_t^2 - 1] = 0$, so $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] = 0$: the score is a martingale difference.
- (b) Step 3: $\partial z_t^2 / \partial \theta = -z_t^2 g_t$, so $\frac{\partial s_t}{\partial \theta^\top} = \frac{1}{2}(z_t^2 - 1) \frac{\partial g_t}{\partial \theta^\top} - \frac{1}{2} z_t^2 g_t g_t^\top$.
- Step 4: the first term has conditional mean 0 and $\mathbb{E} z_t^2 = 1$, hence $A = \frac{1}{2} \mathbb{E}[g_t g_t^\top]$.
- Step 5: $\mathbb{E}(z_t^2 - 1)^2 = \kappa - 2 + 1 = \kappa - 1$, hence $B = \frac{1}{4}(\kappa - 1) \mathbb{E}[g_t g_t^\top] = \frac{\kappa - 1}{2} A$.

A1. The QMLE Sandwich for GARCH(1,1) (Solution 2/2) [Solved]

- Step 6: $A^{-1}BA^{-1} = \frac{\kappa-1}{2}A^{-1}AA^{-1} = \frac{\kappa-1}{2}A^{-1}$; it equals the Hessian-only matrix A^{-1} only if $\kappa = 3$, the kurtosis of the Normal distribution.
- (c) Step 7: $\kappa = 3(\hat{\nu} - 2)/(\hat{\nu} - 4) = 3 \cdot 3.011/1.011 = 8.93$.
- Step 8: variances $\times (\kappa - 1)/2 = 3.97$; standard errors $\times \sqrt{3.97} = 1.99$.
- Interpretation: with BET-like tails the Hessian-only standard errors are about half the correct ones, so their t -statistics are about twice too large; report the sandwich [Bollerslev & Wooldridge \(1992\)](#).
- With $\nu \leq 4$, $\kappa = \infty$, B does not exist and the QMLE is no longer $\sqrt{T}$ -Normal: the usual intervals lose their justification.

A2. The Delta Method for the Half-Life [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Half-life of a volatility shock $h(p) = \ln 0.5 / \ln p$, where $p = \alpha + \beta$ is the persistence of a GARCH(1,1).
- ▶ BET estimates of B3 with robust covariance: $\hat{p} = 0.9907$, $\text{se}(\hat{p}) = 0.0084$; $\hat{p}$ is asymptotically Normal (A1); level 95%.

□ Tasks

- ▶ (a) Derive $h'(p)$ and write the delta-method standard error $\text{se}(\hat{h}) = |h'(\hat{p})| \text{se}(\hat{p})$.
- ▶ (b) Compute $\hat{h}$, $|h'(\hat{p})|$, $\text{se}(\hat{h})$ and the Wald interval $\hat{h} \pm 1.96 \text{se}(\hat{h})$.
- ▶ (c) Using $\ln p \approx -(1-p)$, show that $|h'(p)| \approx \ln 2 / (1-p)^2$ as $p \rightarrow 1$; evaluate the exact and the approximate slope at $p = 0.99, 0.995$ and 0.999 .

□ Report

- ▶ $\hat{h}$ with its standard error and Wald interval; a table of $|h'(p)|$, exact and approximate.
- ▶ Model: A1

□ Interpretation question

- ▶ Why is a Normal interval for the half-life meaningless here, and what should you report instead?

A3. GARCH(1,1) Forecast for BET (1/2) [Solved]

Task

▣ Data and assumptions

- ▶ BET daily log returns in % (data/market/BET.csv), 6 Jan 2000 – 18 Sep 2026; GARCH(1,1)- t on the full sample: $\omega = 0.0448$, $\alpha = 0.1916$, $\beta = 0.7991$.
- ▶ Tomorrow's variance, the one-step forecast $\sigma_{T+1}^2 = \omega + \alpha \varepsilon_T^2 + \beta \sigma_T^2 = 3.11$ ($\sigma_{T+1} = 1.76\%$); forecasts $\sigma_{T+h}^2 = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{T+1}^2 - \bar{\sigma}^2)$; 252 trading days a year.

▣ Tasks

- ▶ (a) Compute the persistence $\alpha + \beta$, the long-run variance $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$, the daily volatility $\bar{\sigma}$ and the annualised volatility $\sqrt{252} \bar{\sigma}$.
- ▶ (b) Compute the half-life of a shock $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$.
- ▶ (c) Forecast the variance on day 10, σ_{T+10}^2 .
- ▶ (d) Compute the variance of the 10-day return $\sum_{h=1}^{10} \sigma_{T+h}^2$ with the closed form of the geometric sum, the 10-day volatility, and compare it with the square-root-of-time rule $\sqrt{10} \sigma_{T+1}$.

A3. GARCH(1,1) Forecast for BET (2/2) [Solved]

Task

☐ Report

- ▶ Every number of (a)–(d), each with the substituted formula.

☐ Interpretation question

- ▶ Why does the forecast move towards the long-run level, and when does the square-root-of-time rule understate or overstate the 10-day risk?

A3. GARCH(1,1) Forecast for BET (Solution 1/2) [Solved]

- (a) Step 1: $\alpha + \beta = 0.9907$, so $1 - \alpha - \beta = 0.0093$.
- Step 2: $\bar{\sigma}^2 = 0.0448/0.0093 = 4.82$, i.e. $\bar{\sigma} = 2.19\%$ a day.
- Step 3: $\sqrt{252 \cdot 4.82} = \sqrt{1214.6} = 34.9\%$ a year.
- (b) Step 4: $h_{1/2} = \ln 0.5 / \ln 0.9907 = 74.2$ trading days, about 3.5 months of 21 trading days.
- Reading: a shock to variance loses half of its effect in about 74 trading days, so BET volatility is very persistent.

A3. GARCH(1,1) Forecast for BET (Solution 2/2) [Solved]

- (c) Step 5: gap to the long run $\sigma_{T+1}^2 - \bar{\sigma}^2 = 3.11 - 4.82 = -1.71$; $(\alpha + \beta)^9 = 0.919$.
- Step 6: $\sigma_{T+10}^2 = 4.82 + 0.919 \cdot (-1.71) = 3.25$: today is below the long run, so the forecast rises.
- (d) Step 7: $\sum_{h=1}^{10} \sigma_{T+h}^2 = 10\bar{\sigma}^2 + (\sigma_{T+1}^2 - \bar{\sigma}^2) \frac{1 - (\alpha + \beta)^{10}}{1 - \alpha - \beta} = 10 \cdot 4.82 + (-1.71) \cdot 9.592 = 31.80$.
- Step 8: 10-day volatility $\sqrt{31.80} = 5.64\%$, against $\sqrt{10} \cdot 1.76 = 5.57\%$ by the square-root-of-time rule.
- Interpretation: variance reverts to $\bar{\sigma}^2$ at rate $\alpha + \beta$; the square-root-of-time rule understates the 10-day risk when volatility is below its long-run level (as now) and overstates it after a crisis spike.

A4. GARCH Forecast with Given Parameters [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ A GARCH(1,1) for daily returns in % with $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$; tomorrow's variance $\sigma_{T+1}^2 = 4$ ($\sigma_{T+1} = 2\%$).
- ▶ Forecast formula and geometric sum as in A3; 252 trading days a year.

▣ Tasks

- ▶ (a) Compute the persistence, the long-run variance $\bar{\sigma}^2$ and the annualised long-run volatility.
- ▶ (b) Compute the half-life of a shock.
- ▶ (c) Forecast the variance on day 5, σ_{T+5}^2 .
- ▶ (d) Compute the variance of the 5-day return and the 5-day volatility, and compare them with $5\sigma_{T+1}^2 = 20$ and $\sqrt{5}\sigma_{T+1}$.

▣ Report

- ▶ Every number of (a)–(d) with the substituted formula.
- ▶ Model: A3

▣ Interpretation question

- ▶ Is the 5-day volatility above or below $\sqrt{5}\sigma_{T+1}$, and why?

A5. Normal VaR 1% and ES 2.5% [Solved]

Task

▣ Data and assumptions

- ▶ A position of 1 million RON in the BET; one-day return $r \sim N(\mu, \sigma^2)$ with $\mu = 0.05\%$, $\sigma = 1.4\%$; loss $L = -r$.
- ▶ Definitions: $\text{VaR}_\alpha = -q_\alpha(r)$, $\text{ES}_\alpha = -\mathbb{E}[r \mid r \leq q_\alpha(r)]$; for the Normal distribution $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$ and $\text{ES}_\alpha = -\mu + \sigma \varphi(z_\alpha)/\alpha$, with φ the standard Normal density.

▣ Tasks

- ▶ (a) Find $z_{0.01}$ and compute $\text{VaR}_{1\%}$ in % and in RON.
- ▶ (b) Find $z_{0.025}$ and $\varphi(z_{0.025})$ and compute $\text{ES}_{2.5\%}$ in % and in RON.
- ▶ (c) Compute the ratio $\text{ES}_{2.5\%}/\text{VaR}_{1\%}$.

▣ Report

- ▶ Both measures in % and in RON, and their ratio.

▣ Interpretation question

- ▶ Why did the Basel Committee replace VaR 1% by ES 2.5%, if the two numbers are almost equal under the Normal distribution?

A5. Normal VaR 1% and ES 2.5% (Solution) [Solved]

- (a) Step 1: $z_{0.01} = -2.326$, so $\sigma z_{0.01} = 1.4 \cdot (-2.326) = -3.257$.
- Step 2: $\text{VaR}_{1\%} = -(0.05 - 3.257) = 3.207\%$, i.e. $3.207\% \cdot 10^6 = 32\,069$ RON.
- (b) Step 3: $z_{0.025} = -1.960$, $\varphi(z_{0.025}) = 0.0584$, $\varphi(z_{0.025})/0.025 = 2.338$.
- Step 4: $\text{ES}_{2.5\%} = -0.05 + 1.4 \cdot 2.338 = -0.05 + 3.273 = 3.223\%$, i.e. $32\,229$ RON.
- (c) Step 5: $\text{ES}_{2.5\%}/\text{VaR}_{1\%} = 3.223/3.207 = 1.005$: under the Normal distribution the two numbers are almost equal.
- Interpretation: the switch is not about the level under the Normal distribution; ES looks inside the tail and is subadditive, while VaR only marks the entrance of the tail (Chapter 7); with heavy tails ES rises more (A6).

A6. Student- t VaR 1% and ES 2.5% [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Same position, $\mu = 0.05\%$ and $\sigma = 1.4\%$ as in A5, but $r = \mu + s T_\nu$, T_ν Student- t with $\nu = 5$, scale $s = \sigma \sqrt{(\nu - 2)/\nu}$ so that the variance is σ^2 .
- ▶ Formulas: $\text{VaR}_\alpha = -(\mu + s q_\alpha)$ with $q_\alpha = t_\nu^{-1}(\alpha)$; $\text{ES}_\alpha = -\mu + s \frac{f_\nu(q_\alpha)}{\alpha} \frac{\nu + q_\alpha^2}{\nu - 1}$, with f_ν the Student- t density.

□ Tasks

- ▶ (a) Compute the scale s .
- ▶ (b) Find $t_5^{-1}(0.01)$ and compute $\text{VaR}_{1\%}$ in % and in RON.
- ▶ (c) Find $q = t_5^{-1}(0.025)$, compute the factors $f_5(q)/0.025$ and $(5 + q^2)/4$, then $\text{ES}_{2.5\%}$ in % and in RON.
- ▶ (d) Compute the ratios Student- t /Normal for VaR and for ES (Normal values from A5), and the ratio ES/VaR.

□ Report

- ▶ Both measures in % and RON, and the three ratios.
- ▶ Model: A5

□ Interpretation question

- ▶ Which number changes more when the tails become heavy, VaR or ES, and what does this mean for capital?

A7. The Power of the Kupiec Test [Solved]

Task

▣ Data and assumptions

- ▶ A VaR 1% model is backtested over T days; the number of breaches is $X \sim \text{Bin}(T, \pi)$; $H_0: \pi = 0.01$.
- ▶ Kupiec statistic $LR_{uc}(x) = -2[(T - x) \ln 0.99 + x \ln 0.01 - (T - x) \ln(1 - x/T) - x \ln(x/T)]$ (with $0 \ln 0 = 0$); reject at 5% if $LR_{uc} > \chi^2_{1;0.95} = 3.84$ Kupiec (1995).

▣ Tasks

- ▶ (a) For $T = 250$, compute $LR_{uc}(x)$ for $x = 0, 1, 2, \dots$ and find the rejection region R ; repeat for $T = 1000$.
- ▶ (b) Compute the exact size $P(X \in R \mid \pi = 1\%)$ for both T .
- ▶ (c) Compute the exact power $P(X \in R \mid \pi = 2\%)$ for both T .

▣ Report

- ▶ The two rejection regions, the exact sizes and the exact powers.

▣ Interpretation question

- ▶ How often does one year of data catch a model whose true breach rate is twice the nominal one, and why does Basel add the traffic-light multiplier?

A7. The Power of the Kupiec Test (Solution 1/2) [Solved]

- (a) Step 1 ($T = 250$, expected 2.5 breaches): $LR_{uc}(0) = -2 \cdot 250 \ln 0.99 = 5.03 > 3.84$, so $x = 0$ is rejected; $LR_{uc}(1) = 1.18$.
- Step 2: $LR_{uc}(6) = 3.56 < 3.84$ and $LR_{uc}(7) = 5.50 > 3.84$, so $R = \{0\} \cup \{x \geq 7\}$.
- Step 3 ($T = 1\,000$, expected 10 breaches): the same scan gives $R = \{x \leq 4\} \cup \{x \geq 17\}$.
- (b) Step 4: at $T = 250$, size = $P(X = 0) + P(X \geq 7) = 0.0811 + 0.0137 = 0.095$ under $\pi = 1\%$; at $T = 1\,000$ the size is 0.055.
- Reading: X is discrete, so the exact size is not 5%; with few expected breaches it can be almost twice the nominal level.

A7. The Power of the Kupiec Test (Solution 2/2) [Solved]

- (c) Step 5: at $T = 250$, $\text{power} = P(X = 0) + P(X \geq 7) = 0.0064 + 0.2363 = 0.243$ under $\pi = 2\%$.
- Step 6: at $T = 1000$, $P(X \leq 4) + P(X \geq 17) = 0.782$ under $\pi = 2\%$.

	$T = 250$	$T = 1\,000$
Rejection region	$x = 0 \text{ or } x \geq 7$	$x \leq 4 \text{ or } x \geq 17$
Exact size ($\pi = 1\%$)	0.095	0.055
Power ($\pi = 2\%$)	0.243	0.782

- Interpretation: one year of data rejects a model with twice the breach rate only about one time in four; four years are needed for a power near 80%; that is why Basel adds the traffic-light multiplier instead of relying on the test alone.

A8. The Power of the Christoffersen Test [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Breaches I_t follow a two-state Markov chain with the correct unconditional rate $\pi = 1\%$ but clustered: $\pi_{11} = P(I_t = 1 \mid I_{t-1} = 1) = 0.20$ (independence would mean $\pi_{11} = 1\%$).
- ▶ Christoffersen independence test $LR_{ind} \sim \chi^2(1)$, rejection at 5% if $LR_{ind} > 3.84$ Christoffersen (1998); a path without any breach counts as a non-rejection.

□ Tasks

- ▶ (a) Derive $\pi_{01} = P(I_t = 1 \mid I_{t-1} = 0)$ from the stationarity condition $\pi = \pi_{01}/(1 - \pi_{11} + \pi_{01})$.
- ▶ (b) Simulate 20 000 paths of $T = 250$ days (first day a breach with probability π), compute LR_{ind} on each path and estimate the power as the share of rejections, with its Monte Carlo standard error.
- ▶ (c) Repeat (b) for $T = 1\,000$ days with 5 000 paths.
- ▶ (d) Compute the expected number of consecutive breach pairs $(T - 1)\pi\pi_{11}$ for both T .

□ Report

- ▶ π_{01} , both powers with the Monte Carlo standard error, the expected numbers of pairs.
- ▶ Model: A7

□ Interpretation question

- ▶ Why is independence almost untestable with one year of data?

A9. Holm and Benjamini–Hochberg [Solved]

Task

□ Data and assumptions

- ▶ Ten trading strategies, each tested against a zero excess return; sorted p -values $p_{(1)}, \dots, p_{(10)}$: 0.001, 0.004, 0.012, 0.019, 0.028, 0.041, 0.09, 0.20, 0.46, 0.73; $m = 10$; target level 5%.
- ▶ FWER: probability of at least one false rejection; $\text{FDR} = \mathbb{E}[V / \max(R, 1)]$, the expected share of false rejections V among all R rejections (0 when nothing is rejected).

□ Tasks

- ▶ (a) Count the rejections without correction ($p \leq 0.05$) and with Bonferroni ($p \leq 0.05/m$).
- ▶ (b) Apply Holm at FWER 5% **Holm (1979)**: compare $p_{(i)}$ with $0.05/(m - i + 1)$ for $i = 1, 2, \dots$ and stop at the first $p_{(i)}$ above its threshold.
- ▶ (c) Apply Benjamini–Hochberg at FDR 5% **Benjamini & Hochberg (1995)**: find the largest k with $p_{(k)} \leq 0.05k/m$ and reject $H_{(1)}, \dots, H_{(k)}$.

□ Report

- ▶ A table with $p_{(i)}$ and both thresholds, and the number of rejections of each procedure.

□ Interpretation question

- ▶ What does each procedure promise, and which one would you use to screen trading strategies?

A9. Holm and Benjamini–Hochberg (Solution 1/2) [Solved]

i	1	2	3	4	5	6	7	8	9	10
$p_{(i)}$	0.001	0.004	0.012	0.019	0.028	0.041	0.09	0.20	0.46	0.73
Holm	0.0050	0.0056	0.0063	0.0071	0.0083	0.0100	0.0125	0.0167	0.0250	0.0500
BH	0.005	0.010	0.015	0.020	0.025	0.030	0.035	0.040	0.045	0.050

- (a) Unadjusted: 6 p -values below 5%; Bonferroni ($p \leq 0.005$): 2 rejections.
- (b) Holm, step down: $0.001 \leq 0.0050$ and $0.004 \leq 0.0056$ pass; $0.012 > 0.0063$ stops the procedure: 2 rejections.

A9. Holm and Benjamini–Hochberg (Solution 2/2) [Solved]

- (c) BH, step up from the largest p : $0.041 > 0.030$, $0.028 > 0.025$, but $0.019 \leq 0.020$, so $k = 4$: 4 rejections (also $0.012 \leq 0.015$ is not needed, since BH rejects all $H_{(i)}$ with $i \leq k$).
- Summary: none 6, Bonferroni 2, Holm 2, BH 4.
- Holm promises: probability of any false discovery $\leq 5\%$ (FWER), uniformly more powerful than Bonferroni.
- BH promises: expected share of false discoveries among the declared ones $\leq 5\%$ (FDR) when the p -values are independent or positively regression dependent on the true nulls [Benjamini & Yekutieli \(2001\)](#); it rejects more.
- Interpretation: for screening many candidate strategies BH is the usual choice (a few false leads are acceptable); before committing capital to one strategy, FWER control (Holm) is the safer promise.

A10. Hurdles for One Hundred Factors [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ 100 candidate factors, each tested with a two-sided t -statistic that is approximately $N(0, 1)$ under H_0 ; target level 5%.
- ▶ Bonferroni and BH as in A9; under arbitrary dependence the Benjamini–Yekutieli procedure replaces $0.05k/m$ by $0.05k/[m c(m)]$, $c(m) = \sum_{i=1}^m 1/i$ [Benjamini & Yekutieli \(2001\)](#).

□ Tasks

- ▶ (a) Compute the Bonferroni per-test level at FWER 5% and the hurdle $|t| > z_{1-0.05/(2m)}$.
- ▶ (b) Compute the BH hurdle for $|t|$ when exactly 10 factors are declared and when only 1 is declared.
- ▶ (c) Compute $c(100)$ and the Benjamini–Yekutieli hurdle when 10 factors are declared.
- ▶ (d) Compare the hurdles with $|t| > 3$ proposed by [Harvey, Liu & Zhu \(2016\)](#).

□ Report

- ▶ The per-test levels and the four $|t|$ hurdles.
- ▶ Model: A9

□ Interpretation question

- ▶ Why does the BH hurdle depend on how many factors are declared, and is BH still valid when factors are positively correlated?

A11. Why Kurtosis Fails with a Tail Index below 4 [Solved]

Task

▣ Data and assumptions

- ▶ $|X|$ has a Pareto-type tail: $P(|X| > x) = Cx^{-\alpha}$ for $x \geq x_0 > 0$, $C > 0$; α is the tail index.
- ▶ Simulation design: Student- t with 3 degrees of freedom (tail index 3, finite variance), sample sizes $T = 10^3, 10^4, 10^5$, 200 samples of each size.

▣ Tasks

- ▶ (a) Use $\mathbb{E}|X|^4 = \int_0^\infty 4x^3 P(|X| > x) dx$ to show that $\mathbb{E}|X|^4 = \infty$ when $\alpha \leq 4$.
- ▶ (b) Simulate the samples, compute the sample excess kurtosis of each, and report its median and 90% quantile for each T .
- ▶ (c) Explain what a bootstrap interval for the kurtosis estimates in this case [Athreya \(1987\)](#).

▣ Report

- ▶ The proof of (a) and a table of the median and 90% quantile for the three sample sizes.

▣ Interpretation question

- ▶ What should you report instead of the kurtosis when the tail index is below 4?

A11. Why Kurtosis Fails with a Tail Index below 4 (Solution 1/2) [Solved]

- (a) Step 1: for $x \geq x_0$ the integrand is $4x^3 \cdot Cx^{-\alpha} = 4Cx^{3-\alpha}$, and it is non-negative everywhere.
- Step 2: $\mathbb{E}|X|^4 \geq 4C \int_{x_0}^{\infty} x^{3-\alpha} dx$, and this integral diverges when $3 - \alpha \geq -1$, i.e. $\alpha \leq 4$.
- Step 3: for $2 < \alpha \leq 4$ the variance is finite but $\mathbb{E}X^4 = \infty$, so the population kurtosis $\mathbb{E}X^4/(\mathbb{E}X^2)^2$ is infinite and the sample kurtosis has no finite limit.
- (b) Steps: draw 200 samples of each size, compute the sample excess kurtosis, take the median and the 90% quantile across the 200 values.

T	10^3	10^4	10^5
Median excess kurtosis	14.5	29.4	73.7
90% quantile	56.9	134.0	376.4

A11. Why Kurtosis Fails with a Tail Index below 4 (Solution 2/2) [Solved]

- Reading: the median grows with T and the 90% quantile grows faster: the statistic does not settle down, it grows without limit.
- (c) There is no population value for an interval to cover, and the bootstrap of such moments is inconsistent [Athreya \(1987\)](#): a bootstrap interval for the kurtosis has no coverage guarantee.
- Interpretation: when the tail index may be below 4, report a tail index with its interval (Hill with block bootstrap, B1) and not the kurtosis; a large sample kurtosis says only that the tails are heavy, not how heavy.

A12. Comparing Two Tail Indices (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Hill estimator on the k largest losses: $\hat{\alpha}^{-1} = \frac{1}{k} \sum_{i=1}^k \ln L_{(i)} - \ln L_{(k+1)}$, $k = 5\%$ of the loss days; $\text{se}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$ for i.i.d. data [Hill \(1975\)](#).
- ▶ From B1–B2: BET $\hat{\alpha} = 2.39$ ($k = 153$), Bitcoin $\hat{\alpha} = 2.74$ ($k = 104$); block-bootstrap standard errors (999 resamples, 20-day blocks): BET 0.239, Bitcoin 0.225; simplifying assumption: the two estimates are independent (the samples overlap in time, so common shocks would add $-2 \text{Cov}(\hat{\alpha}_{\text{BTC}}, \hat{\alpha}_{\text{BET}})$ to the variance of the difference); level 5%.

□ Tasks

- ▶ (a) Compute the i.i.d. standard errors of both estimates.
- ▶ (b) Test $H_0: \alpha_{\text{BTC}} = \alpha_{\text{BET}}$ with $z = (\hat{\alpha}_{\text{BTC}} - \hat{\alpha}_{\text{BET}}) / \sqrt{\text{se}_{\text{BTC}}^2 + \text{se}_{\text{BET}}^2}$, two-sided, with the i.i.d. standard errors.
- ▶ (c) Repeat the test with the block-bootstrap standard errors.

A12. Comparing Two Tail Indices (2/2) [Proposed]

Solution discussed in class

Task

□ Report

- ▶ All standard errors, both z -statistics, p -values and decisions.
- ▶ Model: A11, B1

□ Interpretation question

- ▶ Why can the i.i.d. formula misstate the uncertainty for returns, and does the conclusion depend on it?

Data for Part B

- ▣ BET index, official daily closing values (`data/market/BET.csv`), 6 Jan 2000 – 18 Sep 2026: 6 670 daily log returns in %
- ▣ Bitcoin in USD (`data/market/BTC-USD.CC.csv`), every calendar day, 18 Sep 2014 – 18 Sep 2026: 4 384 daily log returns
- ▣ SPY (S&P 500 ETF, dividends reinvested, `data/market/SPY.US.csv`) for B8, weekdays only
- ▣ One-day VaR 1% forecasts: window of 1000 days for GARCH- t and FHS (re-estimated every 21 days), 500 days for HS and Normal
- ▣ All code: the Quantlet MFM_ch19_seminar and the seminar notebook

B1. Stylised Facts of BET with Robust Inference (1/2) [Solved]

Task

□ Data and assumptions

- ▶ BET official daily closes (data/market/BET.csv, column close), 6 Jan 2000 – 18 Sep 2026: $T = 6670$ log returns $r_t = 100 \ln(P_t/P_{t-1})$ in %; losses $L_t = -r_t$.
- ▶ Moving-block bootstrap: 999 resamples built from 20-day blocks [Künsch \(1989\)](#); all tests at the 5% level.

□ Tasks

- ▶ (a) Compute the standard deviation, the skewness and the excess kurtosis of r_t .
- ▶ (b) Estimate the Hill tail index of losses [Hill \(1975\)](#) with $k = 5\%$ of the loss days; compute the asymptotic 95% CI $\hat{\alpha}(1 \pm 1.96/\sqrt{k})$ and the 95% percentile CI of the block bootstrap.
- ▶ (c) Compute the Ljung–Box $Q(10)$ on r_t and on r_t^2 , and the robust $Q^*(10) = T \sum_{k=1}^{10} \hat{\rho}_k^2 / \hat{\tau}_k$, $\hat{\tau}_k = \overline{x_t^2 x_{t-k}^2} / \hat{\gamma}_0^2$, with $x_t = r_t - \bar{r}$ and $\hat{\gamma}_0 = \overline{x_t^2}$ [Lobato, Nankervis & Savin \(2001\)](#); compare each with $\chi_{10,0.95}^2 = 18.31$; the χ^2 calibration of Q^* assumes finite fourth moments.
- ▶ (d) Count the days with $|r_t - \bar{r}| > 4s$ and compare with the number expected under the Normal distribution, $T \cdot 2\Phi(-4)$.

B1. Stylised Facts of BET with Robust Inference (2/2) [Solved]

Task

□ Report

- ▶ The statistics of (a), $\hat{\alpha}$ with both intervals, the three portmanteau statistics with p -values and decisions, the count of (d).

□ Interpretation questions

- ▶ Why is the block interval for the tail index wider than the asymptotic one?
- ▶ Does the autocorrelation of r_t survive the robust test, and how far can you trust its p -value given the tail index? What could explain it?

B1. Stylised Facts of BET with Robust Inference (Solution 1/2) [Solved]

- (a) $s = 1.40\%$, skewness -0.65 , excess kurtosis 11.1 : fat, left-skewed tails.
- (b) Step 1: sort the losses; $k = 153$ (5% of the loss days); $\hat{\alpha} = 2.39$.
- Step 2 (asymptotic): $2.39(1 \pm 1.96/\sqrt{153}) = [2.01; 2.77]$, standard error 0.193 .
- Step 3 (block bootstrap): rebuild 999 series from random 20-day blocks, re-estimate $\hat{\alpha}$ with the same k , take the 2.5% and 97.5% percentiles: $[2.15; 3.09]$, standard error 0.239 .
- Reading: both intervals lie below 4, so the fourth moment is infinite and the sample kurtosis (11.1) is not reported as an estimate (A11).

B1. Stylised Facts of BET with Robust Inference (Solution 2/2) [Solved]

- (c) $Q(10)$ on r_t : 108.0 ($p < 0.001$) > 18.31 : reject no autocorrelation; robust $Q^*(10) = 24.4$ ($p = 0.007$): still reject, but 4.4 times smaller.
- $Q(10)$ on r_t^2 : 2440 ($p < 0.001$): strong volatility clustering, read descriptively, since with an infinite fourth moment r_t^2 has no finite variance; first-order autocorrelation of r_t : 0.109 .
- (d) 51 days beyond ± 4 standard deviations, against $6\,670 \cdot 2\Phi(-4) = 0.42$ expected under the Normal distribution.
- Interpretation 1: extremes cluster in crises; blocks keep the clusters, so the block interval accounts for the dependence that the asymptotic formula ignores.
- Interpretation 2: the robust test still rejects, but with $\hat{\alpha} < 4$ its χ^2 calibration is not guaranteed: sample autocorrelations of heavy-tailed GARCH-type returns converge more slowly and to non-Normal limits [Davis & Mikosch \(1998\)](#); read it as evidence of a small autocorrelation; stale prices of less liquid stocks (Chapters 2, 10) are a plausible explanation that needs its own test.

B2. Stylised Facts of Bitcoin (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Bitcoin in USD (data/market/BTC-USD.CC.csv, column close), every calendar day, 18 Sep 2014 – 18 Sep 2026:
 $T = 4\,384$ log returns in %.
- ▶ Same settings as B1: $k = 5\%$ of the loss days, 999 block-bootstrap resamples with 20-day blocks, 10 lags, level 5%; BET results from B1.

□ Tasks

- ▶ (a) Compute the standard deviation, skewness and excess kurtosis.
- ▶ (b) Estimate the Hill tail index with the asymptotic and the block-bootstrap 95% CI.
- ▶ (c) Compute $Q(10)$ on r_t and r_t^2 and the robust $Q^*(10)$, and decide at 5%.
- ▶ (d) Count the days beyond ± 4 standard deviations and compare with the Normal distribution.
- ▶ (e) Test $H_0: \alpha_{\text{BTC}} = \alpha_{\text{BET}}$ with $z = (\hat{\alpha}_{\text{BTC}} - \hat{\alpha}_{\text{BET}}) / \sqrt{\text{se}_{\text{BTC}}^2 + \text{se}_{\text{BET}}^2}$ using the block-bootstrap standard errors (A12).

B2. Stylised Facts of Bitcoin (2/2) [Proposed]

Solution discussed in class

Task

□ Report

- ▶ A table Bitcoin vs BET with (a)–(d); the test of (e) with z , p and decision.
- ▶ Model: B1

□ Interpretation questions

- ▶ Are Bitcoin's tails heavier than the BET's, or is Bitcoin only more volatile?
- ▶ Does the linear autocorrelation of Bitcoin survive the robust test?

B3. GARCH(1,1)- t for BET: an Interval at the Boundary (1/2) [Solved]

Task

□ Data and assumptions

- ▶ BET daily log returns in %, 6 Jan 2000 – 18 Sep 2026 (B1); constant mean; $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ with standardised Student- t innovations (ν degrees of freedom); robust standard errors [Bollerslev & Wooldridge \(1992\)](#).
- ▶ Persistence $p = \alpha + \beta$; a finite unconditional variance (covariance stationarity) requires $p < 1$, whereas strict stationarity only needs $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$ [Nelson \(1990\)](#); $\hat{p}$ lies close to the boundary $p = 1$ of the parameter space $p \leq 1$ [Andrews \(1999\)](#); level 95%.

□ Tasks

- ▶ (a) Estimate the model by maximum likelihood; report $\hat{\alpha}$, $\hat{\beta}$, $\hat{\nu}$ with robust standard errors and $\hat{p}$.
- ▶ (b) Compute $\text{se}(\hat{p}) = \sqrt{\widehat{\text{Var}}(\hat{\alpha}) + \widehat{\text{Var}}(\hat{\beta}) + 2\widehat{\text{Cov}}(\hat{\alpha}, \hat{\beta})}$ and the Wald CI $\hat{p} \pm 1.96 \text{se}(\hat{p})$.
- ▶ (c) Compute the profile-likelihood CI: for each p on the grid 0.960, 0.961, $\dots$, 0.998, 0.999, 0.9995, 0.9999 and 1 (IGARCH), maximise the log-likelihood over μ , ω , ν and $s = \alpha/p$; keep every $p < 1$ with $2(\ell_{\max} - \ell(p)) \leq 3.84$, and keep $p = 1$ if the statistic is ≤ 2.71 , the 5% critical value of its boundary distribution $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ [Andrews \(1999\)](#).
- ▶ (d) Transform the estimate and both intervals into half-lives $\ln 0.5 / \ln p$.
- ▶ (e) Compare the 1% quantile of the standardised Student- t with $\hat{\nu}$ degrees of freedom, $t_{\hat{\nu}}^{-1}(0.01)\sqrt{(\hat{\nu} - 2)/\hat{\nu}}$, with $z_{0.01}$.

B3. GARCH(1,1)- t for BET: an Interval at the Boundary (2/2) [Solved]

Task

☐ Report

- ▶ Estimates with standard errors; both CIs for p and for the half-life; the profile statistic at the edge of the interval; the two quantiles.

☐ Interpretation questions

- ▶ Which interval can you report, and why is the other one wrong?
- ▶ What does $\hat{\nu} \approx 5$ mean for VaR?

B3. GARCH(1,1)- t for BET: an Interval at the Boundary (Solution 1/2) [Solved]

- (a) $\hat{\alpha} = 0.192$ (se 0.025), $\hat{\beta} = 0.799$ (se 0.024), $\hat{\nu} = 5.01$ (se 0.31); $\hat{\rho} = 0.9907$.
- (b) With the robust covariance, $\text{se}(\hat{\rho}) = 0.0084$ (the negative covariance of $\hat{\alpha}$ and $\hat{\beta}$ makes it smaller than either standard error).
- Wald CI: $0.9907 \pm 1.96 \cdot 0.0084 = [0.974; 1.007]$: the upper end is above 1, outside the parameter space $\rho \leq 1$.
- Why it is wrong: at the boundary the distribution of $\hat{\rho}$ is not Normal [Andrews \(1999\)](#), and the interval contains values $\rho > 1$, for which the variance explodes and the half-life is not defined.
- (d) Half-life $\ln 0.5 / \ln 0.9907 = 74$ days; from the Wald interval: 26 days at the lower end and no value at the upper end.

B3. GARCH(1,1)- t for BET: an Interval at the Boundary (Solution 2/2) [Solved]

- (c) Profile statistic $2(\ell_{\max} - \ell(p))$: 4.32 at $p = 0.974$ (excluded), 3.83 at $p = 0.975$ (included), 0.99 at $p = 0.9999$ (included); at $p = 1$: $1.01 < 2.71$, boundary p -value 0.16 (included).
- Profile CI $[0.975; 1]$: the likelihood excludes no value from 0.975 up to and including IGARCH, $p = 1$.
- (d) Half-life interval: at least $\ln 0.5 / \ln 0.975 = 27$ days, with no finite upper bound, since $p = 1$ is in the interval.
- (e) $t_{\hat{\nu}}^{-1}(0.01)\sqrt{(\hat{\nu} - 2)/\hat{\nu}} = -2.606$ against $z_{0.01} = -2.326$: for the same σ_t the Student- t VaR 1% is 1.12 times the Normal one (zero mean).
- Interpretation 1: report the profile interval; shocks last at least a month, but the data cannot tell a month from several years, and IGARCH ($p = 1$) is not rejected.
- Interpretation 2: $\hat{\nu} \approx 5$ means heavy tails even conditionally on σ_t ; a Normal GARCH VaR 1% would be too low.

B4. Asymmetry: GJR-GARCH- t for BET [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ BET returns as in B3; GJR-GARCH(1,1)- t : $\sigma_t^2 = \omega + (\alpha + \gamma \mathbb{I}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$; the GARCH(1,1)- t of B3 is the restricted model $\gamma = 0$.
- ▶ $T = 6\,670$; $AIC = -2\ell + 2k$, $BIC = -2\ell + k \ln T$ (k = number of parameters); level 5%.

□ Tasks

- ▶ (a) Estimate the GJR model and report $\hat{\gamma}$, its robust standard error and $t = \hat{\gamma}/\text{se}(\hat{\gamma})$.
- ▶ (b) Compute $LR = 2(\ell_{\text{GJR}} - \ell_{\text{GARCH}})$ and its p -value from $\chi^2(1)$.
- ▶ (c) Compute AIC and BIC of both models and show that $\Delta AIC = 2 - LR$ and $\Delta BIC = \ln T - LR$ (GJR minus GARCH).
- ▶ (d) Compute $(\hat{\alpha} + \hat{\gamma})/\hat{\alpha}$, the impact of a negative shock relative to a positive one.

□ Report

- ▶ $\hat{\gamma}$ with standard error and t ; LR and p ; both criteria for both models; the ratio of (d).
- ▶ Model: B3

□ Interpretation question

- ▶ Do the tests and the criteria agree? Which model would you use for VaR, and how would you decide?

B5. Backtesting VaR 1% and ES 2.5% for BET, 2005–2026 (1/2) [Solved]

Task

□ Data and assumptions

- ▶ BET daily losses $L_t = -r_t$, evaluation 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days; each forecast uses only data up to $t - 1$.
- ▶ Four one-day models [McNeil & Frey \(2000\)](#): HS (500 days), Normal (mean and standard deviation of 500 days), GARCH- t (1000-day window, re-estimated every 21 days), FHS (GARCH- t filter and the empirical quantiles of the standardised residuals); VaR 1% and the pair (VaR 2.5%, ES 2.5%).

□ Tasks

- ▶ (a) Count the breaches $L_t > \text{VaR}_t$, compute the rate, the Kupiec p_{uc} and the Christoffersen independence p_{ind} [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#).
- ▶ (b) Run the DQ test [Engle & Manganelli \(2004\)](#): regress $\text{Hit}_t - 0.01$ on a constant, 4 lagged hits and VaR_t ; $DQ \sim \chi^2(6)$.
- ▶ (c) Compute the average FZ0 score of (VaR 2.5%, ES 2.5%) [Patton, Ziegel & Chen \(2019\)](#), Diebold–Mariano tests with Newey–West variance (9 lags), and the model confidence set at 10% ($T_{\max}$ statistic, 999 block-bootstrap resamples, 20-day blocks) [Hansen, Lunde & Nason \(2011\)](#).
- ▶ (d) Run the comparative ES backtest against HS as the standard model [Nolde & Ziegel \(2017\)](#): with t the Diebold–Mariano statistic of $\text{FZ0}(\text{model}) - \text{FZ0}(\text{HS})$, the zone is red if $t > 1.645$, green if $t < -1.645$, yellow otherwise.

B5. Backtesting VaR 1% and ES 2.5% for BET, 2005–2026 (2/2) [Solved]

Task

□ Report

- ▶ One table with the coverage tests (a)–(b) and one with the scores, the MCS p -values, the ES zones and the average VaR.

□ Interpretation questions

- ▶ Which model meets all the backtesting criteria of this exercise? Which one would a trader prefer, and why might they differ?
- ▶ Do the formal comparisons confirm the backtests?

B5. Backtesting VaR 1% and ES 2.5% for BET, 2005–2026 (Solution 1/2) [Solved]

- (a)–(b) Steps: $\text{Hit}_t = \mathbb{1}\{L_t > \text{VaR}_t\}$; expected breaches $0.01 \cdot 5432 \approx 54$; tests at 5%.

Model	x	Rate (%)	p_{uc}	p_{ind}	p_{DQ}
HS	79	1.45	0.002	<0.001	<0.001
Normal	122	2.25	<0.001	<0.001	<0.001
GARCH- t	80	1.47	0.001	0.864	0.007
FHS	67	1.23	0.095	0.267	0.129

- HS and Normal fail coverage, independence and DQ: they react too slowly and their breaches cluster.
- For GARCH- t independence is not rejected, but it has too many breaches, so it fails coverage and DQ; only FHS is rejected by none of the three tests.

B5. Backtesting VaR 1% and ES 2.5% for BET, 2005–2026 (Solution 2/2) [Solved]

- (c)–(d) Lower FZ0 is better; DM t vs HS < -1.645 means the model beats HS.

Model	FZ0	p_{MCS}	DM t vs HS	ES zone	Mean VaR (%)
HS	1.424	0.00	–	standard	3.66
Normal	1.495	0.00	2.62	red	2.90
GARCH- t	1.175	0.18	–5.75	green	2.95
FHS	1.157	1.00	–5.92	green	3.17

- No statistically detectable FZ0 difference between FHS and GARCH- t (DM $t = 1.39$); together they form the model confidence set {GARCH- t , FHS}; Normal is red, GARCH- t and FHS are green.
- Interpretation 1: FHS is the only model that meets all the stated criteria (coverage, independence, DQ, green ES zone), which is what a regulator's backtest looks at; a trader wants low capital, and the Normal model has the lowest average VaR (2.90% vs FHS 3.17%) while failing every test.
- Interpretation 2: the formal comparisons confirm the backtests and add one thing: GARCH- t fails coverage, but its FZ0 score for (VaR, ES) is not significantly different from that of FHS.

B6. Backtesting VaR and ES for Bitcoin, 2019–2026 (1/2) [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Bitcoin daily losses (data/market/BTC-USD.CC.csv), evaluation 1 Jan 2019 – 18 Sep 2026, $T = 2\,818$ days, every calendar day; each forecast uses only data up to $t - 1$.
- ▶ The four models, windows and tests of B5; Newey–West variance with 8 lags; level 5% (10% for the model confidence set).

▣ Tasks

- ▶ (a) Count the breaches of VaR 1% and compute the rate, p_{uc} and p_{ind} for each model.
- ▶ (b) Run the DQ test with 4 lagged hits and VaR_t .
- ▶ (c) Compute the FZ0 scores, the Diebold–Mariano tests and the model confidence set at 10%.
- ▶ (d) Run the comparative ES backtest against HS.
- ▶ (e) Plot the daily losses and the HS and FHS VaR 1% from 1 January 2022.

B6. Backtesting VaR and ES for Bitcoin, 2019–2026 (2/2) [Proposed]

Solution discussed in class

Task

☐ Report

- ▶ The two tables of B5 for Bitcoin and the chart of (e).
- ▶ Model: B5

☐ Interpretation question

- ▶ Is the ranking of the models the same as for BET? Can the tests separate the models at all?

B7. The Look-Ahead Trap [Solved]

Task

□ Data and assumptions

- ▶ BET losses of B5: evaluation 3 Jan 2005 – 18 Sep 2026, $T = 5432$ days, $0.01T \approx 54$ expected breaches; rolling HS and GARCH- t results from B5.
- ▶ The look-ahead versions use data that were not available at $t - 1$; Kupiec test at 5%.

□ Tasks

- ▶ (a) HS with look-ahead: set every day $\text{VaR}_{1\%} = q_{0.99}(L)$, the empirical quantile of all losses of the evaluation period; count the breaches and compute the rate.
- ▶ (b) GARCH- t with look-ahead: estimate once on the full sample 6 Jan 2000 – 18 Sep 2026, filter $\hat{\sigma}_t$ day by day, set $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t \tilde{t}_\nu^{-1}(0.01))$ with $\tilde{t}$ the standardised Student- t ; count the breaches and run the Kupiec test.
- ▶ (c) Compare both with the rolling versions of B5 in one table.

□ Report

- ▶ A table with breaches, rate and p_{uc} for the four versions.

□ Interpretation question

- ▶ Why does the look-ahead version look better? Could anyone have used it in 2008?

B7. The Look-Ahead Trap (Solution 1/2) [Solved]

- (a) Full-sample HS: $\text{VaR}_{1\%} = 4.24\%$ every day; 55 breaches, rate 1.01%: about 1% by construction, because 1% of the losses of a sample exceed its own 0.99 quantile.
- (b) Full-sample GARCH- t : 71 breaches, rate 1.31%, p_{uc} 0.030: still rejected at 5%, but much closer to 1% than the rolling model.

Model	Parameters	x	Rate (%)	p_{uc}
HS	full sample (look-ahead)	55	1.01	0.93
HS	rolling, 500 days	79	1.45	0.002
GARCH- t	full sample (look-ahead)	71	1.31	0.030
GARCH- t	rolling, 1000 days	80	1.47	0.001

B7. The Look-Ahead Trap (Solution 2/2) [Solved]

- (c) Reading: both look-ahead versions have fewer breaches than their rolling counterparts and look better calibrated.
- Why: the full sample includes 2008, 2020 and every later crash, so the quantile and the parameters have already seen the losses they are tested on.
- Interpretation: no one could have used these numbers in 2008, since they depend on losses from 2008 to 2026; only the rolling result is evidence about a forecasting model.
- Check for your own code: every forecast for day t must be computable from a data file truncated at $t - 1$.

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B8. The Calendar Trap: Bitcoin and SPY (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Bitcoin (data/market/BTC-USD.CC.csv, close, every calendar day) and SPY, the S&P 500 ETF (data/market/SPY.US.csv, adjusted_close, weekdays only); daily log returns in %.
- ▶ Two periods: 1 January 2015 – 31 December 2019 (1 258 common days) and 1 January 2020 – 18 September 2026 (1 687 common days).

□ Tasks

- ▶ (a) Wrong join: compute the log returns of each asset on its own calendar, then keep the common days.
- ▶ (b) Right join: keep the common days of the prices, then compute the log returns (a Monday return then covers Friday to Monday for both assets).
- ▶ (c) For each period and each join compute the Bitcoin–SPY correlation, the standard deviation of Bitcoin returns on all days and on Mondays, and the cumulative Bitcoin return in %, $100 [\exp(\sum_t r_t/100) - 1]$ (with r_t in %).
- ▶ (d) Plot the 250-day rolling correlation from 2016 for both joins.

B8. The Calendar Trap: Bitcoin and SPY (2/2) [Proposed]

Solution discussed in class

Task

☐ Report

- ▶ A table (statistics $\times$ periods $\times$ joins) and the chart of (d).
- ▶ Model: B7

☐ Interpretation question

- ▶ Which statistic is most damaged by the wrong join? Would a portfolio VaR built on it be too low or too high?

Proposal Template (One Page)

Sections

- **Question:** one sentence; H_0 , H_1 and the benchmark to beat
- **Replication target:** the table or figure you will reproduce, with DOI, page and table number
- **Data:** series, owner, period, frequency, cleaning rules
- **Identification:** control market, placebo dates, break test with unknown date, or out-of-sample design

Sections (continued)

- **Tests and power:** the main test, its loss, and the minimum detectable effect at your T
- **Specification grid:** every window, lag, α and sample you will try, and the multiplicity correction
- **Risks:** data problems, too few events, overfitting
- **References:** 10–15, at least 3 from the last five years, each checked to exist (DOI or link)

A Worked Proposal (1/2)

- **Question:** did the tail risk of the BET change after the FTSE reclassification of the BVB as a Secondary Emerging market (21 September 2020)?
- **Hypothesis:** H_0 : same Hill tail index of losses and same VaR 1% breach rates in the six years before and after
- **Data:** BET official closes, 22 Sep 2014 – 18 Sep 2026; control: WIG20 (Poland); duplicate records (rows identical to the previous one) and weekend-dated rows removed; BET and WIG20 joined on common price dates, then returns
- **Identification:** difference-in-differences of Hill indices (BET minus WIG20); placebo dates $\pm$ one year [Andrews \(1993\)](#)
 - ▶ Identifying assumptions: without the event the two tail indices would have changed alike; no spillover of the reclassification to Poland; no concurrent shock specific to one of the two markets
 - ▶ Complementary scale diagnostic: a sup-Wald break test with unknown date on $\ln |r_t|$ (it detects shifts in volatility, not in the tail index)

A Worked Proposal (2/2)

- **Inference:** moving-block bootstrap (999 resamples, 20-day blocks)
 - ▶ Within each period the same blocks of days for BET and WIG20, so the dependence between the two markets is kept; independent draws in the two disjoint periods
- **Power:** minimum detectable difference-in-differences at 80% power and 5% level, $(z_{0.975} + z_{0.80}) se = 2.80 se$ (as in A7 and C1(e))
- **Specification grid:** $k = 2.5\%, 5\%, 10\%$ of the loss days $\times$ blocks of 10, 20, 40 days: 9 tests of the difference-in-differences, Holm correction (A9)
- **Replication target:** not part of this illustration, which is a partial model; your proposal must name the published table or figure, with DOI and location (C1(b))
- **Risks:** the pre-period contains the COVID-19 crash and the 2018 bank-tax day; one event, so no causal claim without the control

C1. Your Team's Proposal (1/2) [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ One market or asset of your choice: a series from `data/market` (list in `data/manifest.csv`) or a public source named by its official owner; daily data unless the question needs another frequency.
- ▶ Use the template of the previous slides; the worked proposal is a partial model (it leaves out the replication target); ideas: Part C of Seminars 0–18.

▣ Tasks

- ▶ (a) Write the question in one sentence, with H_0 , H_1 and the benchmark to beat.
- ▶ (b) Name the published table or figure you will replicate first, with DOI, page and table number.
- ▶ (c) Specify the data: series, owner, period, frequency and cleaning rules.
- ▶ (d) Describe the identification: control market, placebo dates, break test with unknown date, or out-of-sample design.
- ▶ (e) State the main test and its loss function, and compute the smallest effect your design detects with 80% power at your T (as in A7).
- ▶ (f) List the specification grid (windows, lags, α levels, samples), count the tests and choose the multiplicity correction (A9–A10).

C1. Your Team's Proposal (2/2) [Proposed]

Solution discussed in class

Task

▣ Report

- ▶ The filled template on one page, with 10–15 checked references, within one week.
- ▶ Model: the worked proposal

▣ Checklist before you submit

- ▶ Which published table or figure will you replicate first (step 1 of the project)?
- ▶ Is the question answerable with data you already have, and is the benchmark strong?
- ▶ Is the loss function fixed in advance, and does every forecast use only information available at $t - 1$?
- ▶ Have you counted how many models and tests you will run, and chosen the correction?

C2. Critique an AI-Written Backtest (1/2) [Proposed]

Solution discussed in class

- ▣ **Student prompt:** “bet holds the daily BET closes. Write a backtest of historical-simulation VaR 1% with a 500-day window and a Kupiec test, and give me a reference for the result.”
- ▣ **Answer of the AI assistant**

```
import numpy as np
from scipy.stats import chi2
r = np.log(bet).diff().dropna()
var = -r.rolling(500).quantile(0.01)
hit = (-r > var)[var.notna()]
x, n, p = hit.sum(), hit.size, 0.01
pi = x / n
lr = -2 * (x*np.log(p) + (n-x)*np.log(1-p) - x*np.log(pi) - (n-x)*np.log(1-pi))
print(x, n, chi2.sf(lr, df=2))
```

- ▣ “The breach rate is close to 1% and the Kupiec p -value is above 5%: historical simulation is unbiased for BET, as shown by Kupiec and Christoffersen (2004), *Backtesting historical simulation VaR*, Journal of Risk Model Validation, 8(2), 45–67.”

C2. Critique an AI-Written Backtest (2/2) [Proposed]

Solution discussed in class

- ▣ The answer on the previous slide contains at least **three** errors that often appear in AI-written project code; also assess its conclusion and check how the code behaves when there are zero breaches
 - ▶ Model: B5 and B7 (backtest and the look-ahead trap), A7 (Kupiec test)
- ▣ Your tasks
 - ▶ (a) Find each error: which line or sentence, and why it is wrong
 - ▶ (b) Say how you would detect it without knowing the answer: a timing test, the number of restrictions, a DOI search
 - ▶ (c) Give the corrected code and a correct reference
 - ▶ (d) Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

C3. Project Seed: Nonstandard Errors in a Mini-Multiverse (1/2) [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Context: in [Menkveld et al. \(2024\)](#) 164 teams tested the same hypotheses on the same data; the interquartile range (IQR) of their estimates of the yearly trend in market efficiency, the nonstandard error, was 6.7 percentage points, against a median SE of 2.5% (Table I, Panel C).
- ▶ Data: daily closes of the Euro Stoxx 50 index (data/market/STOXX50E.INDX.csv) and of the BET (data/market/BET.csv), 2002–2018, then up to the last full year; H_0 : “market efficiency has not changed over time” (Sec. 1.A).

▣ Tasks

- ▶ (a) Before any run, commit a plan with the forks of Table V that daily data allow: outliers (none; winsorised or trimmed at the 2.5 and 97.5 percentiles); measure ($|VR(q) - 1|$, variance ratio of q -day to daily returns, $q = 5$ or 21; R^2 of an AR(1) for daily returns); model (linear trend; log difference; relative difference).
- ▶ (b) For each index and each of the $2 \times 3 \times 3 + 3 \times 3 = 27$ paths, compute the annual measure m_t and its average yearly % change with an SE: $100 b / \bar{m}$ from the regression of the measure on a linear trend, the mean of $100 \Delta \ln m_t$, or the mean of $100 (m_t / m_{t-1} - 1)$.
- ▶ (c) Report the median, the IQR (the nonstandard error of your multiverse) and the 10–90 percentile range of the 27 estimates; compare the IQR with $1.35 \times$ the median SE.

C3. Project Seed: Nonstandard Errors in a Mini-Multiverse (2/2) [Proposed]

Solution discussed in class

Task

▣ Tasks

- ▶ (d) Rank the forks by the k -sample Anderson–Darling statistic (`scipy.stats.anderson_ksamp`) of the estimates grouped by the alternative at each fork (Sec. II.C).
- ▶ (e) Test each path's trend against zero at 5% with the Holm correction over the 27 tests, and report how many paths reject with each sign.

▣ Report

- ▶ A table of the 27 estimates with their SEs for each index; the median, IQR and 10–90 range; the Anderson–Darling ranking of the forks; the number of Holm rejections by sign; a box plot of the estimates grouped by the most influential fork.
- ▶ Model: A9 (Holm and BH over a family of tests); the lecture case study

▣ Interpretation questions

- ▶ Is the IQR over 27 paths an estimate of the nonstandard error of 164 human teams? Use footnote 32 of the paper in your answer.
- ▶ If the sign of the trend changes with the model fork, what should your project report, and which rule would you have written into the pre-analysis plan?

Common Pitfalls in Proposals

- ▣ A topic instead of a question (“Bitcoin volatility”)
- ▣ No benchmark, or no test period fixed in advance
- ▣ Data that do not exist at the needed frequency or history
- ▣ Too many markets and models for the time available
- ▣ A single event treated as a causal experiment without a control

Take-Home Assignment (Team Project)

- Submit your one-page proposal (C1) within one week
- Step 1 of the project: choose the published table or figure you will replicate cell by cell and give its source with a DOI; commit the pre-analysis plan before opening the test period
- Open the team repository on GitHub with a README and the two mandatory files
 - ▶ [AI_USE.md](#): which AI tools you used, for what, and what you checked yourselves
 - ▶ [AI_ERRORS.md](#): at least three AI errors you caught
- Log any AI error you catch from now on in [AI_ERRORS.md](#), as in C2

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Statistics of Financial Markets
- ☐  Measuring Statistical Risk

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_19

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